



RV College of Engineering®

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Industrializing AI Fabric Defect Detection using FabricAI Pro Suite

An Interdisciplinary Project Report (XX367P)

Submitted by,

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RV College of Engineering®

In partial fulfillment of the requirements for the degree of
Bachelor of Engineering in respective departments

2024-25



RV College of Engineering®, Bengaluru
(Autonomous institution affiliated to VTU, Belagavi)



CERTIFICATE

Certified that the interdisciplinary project (XX367P) work titled ***Industrializing AI Fabric Defect Detection using FabricAI Pro Suite*** is carried out by **Varshini C B (1RV22XX001)**, **Team Member 2 (1RV22XX002)**, **Team Member 3 (1RV22XX003)**, **Team Member 4 (1RV22XX004)**, **Team Member 5 (1RV22XX005)** and **Team Member 6 (1RV22XX006)** who are bonafide students of RV College of Engineering, Bengaluru, in partial fulfillment of the requirements for the degree of **Bachelor of Engineering** in respective departments during the year 2024-25. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the interdisciplinary project report deposited in the departmental library. The interdisciplinary project report has been approved as it satisfies the academic requirements in respect of interdisciplinary project work prescribed by the institution for the said degree.

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Signature with Date

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- 2.

DECLARATION

We, **Varshini C B, Team Member 2, Team Member 3, Team Member 4, Team Member 5 and Team Member 6** students of sixth semester B.E., RV College of Engineering, Bengaluru, hereby declare that the interdisciplinary project titled '**Industrializing AI Fabric Defect Detection using FabricAI Pro Suite**' has been carried out by us and submitted in partial fulfilment for the award of degree of **Bachelor of Engineering** in respective departments during the year 2024-25.

Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any degree or diploma to any other university.

We also declare that any Intellectual Property Rights generated out of this project carried out at RVCE will be the property of RV College of Engineering, Bengaluru and We will be one of the authors of the same.

Place: Bengaluru

Date:

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Signature

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Lastly, we take this opportunity to thank our family members and friends who provided all the backup support throughout the project work.

ABSTRACT

The industrialization of the FabricAI Pro Suite presents a comprehensive web-based platform designed for real-time anomaly detection in high-speed industrial looms. Fabric defect detection is traditionally constrained by manual inspection limitations and the significant computational latency of classical computer vision models. This report addresses these limitations by detailing the architectural evolution from a local desktop prototype to a decoupled, high-performance web architecture, leveraging WebSocket streaming and GPU-accelerated tensor operations.

The primary objective of this work was to optimize an ensemble AI pipeline—fusing PatchCore, DINO, and a Vision Transformer Autoencoder (ViT-AE)—to deliver high-fidelity spatial anomaly heatmaps. By transitioning the nearest-neighbor clustering in PatchCore from CPU-bound sklearn models to natively batched `torch.cdist` operations on the GPU, we achieved a significant reduction in frame latency, enabling robust 15–30 FPS inference suitable for industrial deployment.

The system incorporates a modern React/Vite frontend with a "Midnight Industrial" aesthetic, presenting a split-view dashboard for real-time comparative analysis between raw camera feeds and AI-processed anomaly maps. Precision bounding box overlays were implemented using contour thresholding techniques to provide intuitive visual feedback on localized defect clusters. The resulting architecture ensures scalable, containerized execution, establishing a robust foundation for modern textile manufacturing environments.

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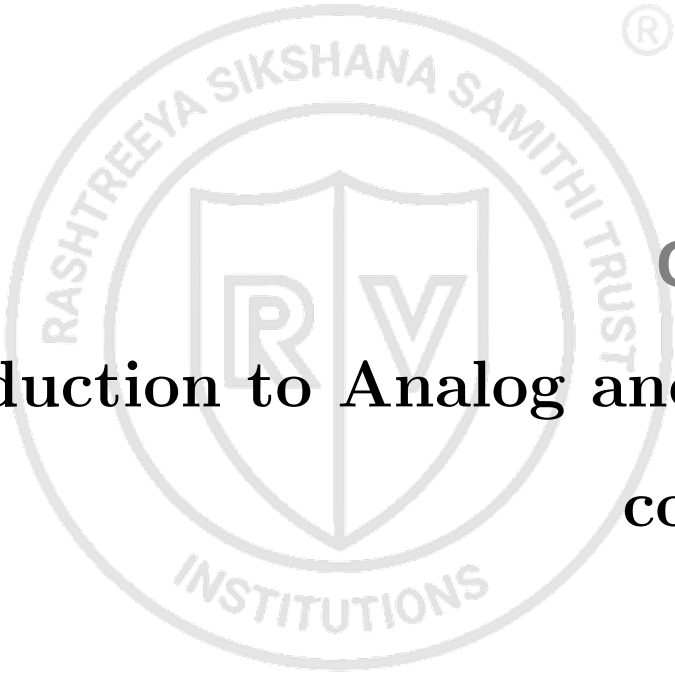
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Chapter 1

Introduction to Analog and Digital converters

CHAPTER 1

INTRODUCTION TO ANALOG AND DIGITAL CONVERTERS

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

1.1 Introduction

The title of the project can be introduced in this section. This section should neatly elaborate the context of the project, the relevance of the area chosen and the title. You can bring a brief history and arrive at the title of the project. Use appropriate number of paragraphs within this section.

You are allowed to use figures or diagrams which can help in introducing the topic acknowledging the source. For example , if you are introducing a particular topic, an appropriate figure can be used. The figure should be referenced in the text as Figure. 1.1



Figure 1.1: Sample picture of universe

These guidelines are provided to formally expose you to the various ethical and technical issues involved in writing up your work and the format you are required to adhere

to while submitting your project report.

1.2 Literature Review

A literature review is a text of a scholarly paper, which includes the current knowledge including substantive findings, as well as theoretical and methodological contributions to a particular topic. Literature reviews are secondary sources, and do not report new or original experimental work. Most often associated with academic-oriented literature, such reviews are found in academic journals, and are not to be confused with book reviews that may also appear in the same publication. Literature reviews are a basis for research in nearly every academic field . A narrow-scope literature review may be included as part of a peer-reviewed journal article presenting new research, serving to situate the current study within the body of the relevant literature and to provide context for the reader. In such a case, the review usually precedes the methodology and results sections of the work.

1.2.1 Sample

The main types of literature reviews are: evaluative, exploratory, and instrumental. A fourth type, the systematic review, is often classified separately, but is essentially a literature review focused on a research question, trying to identify, appraise, select and synthesize all high-quality research evidence and arguments relevant to that question. A meta-analysis is typically a systematic review using statistical methods to effectively combine the data used on all selected studies to produce a more reliable result.

Review types

The main types of literature reviews are: evaluative, exploratory, and instrumental. A fourth type, the systematic review, is often classified separately, but is essentially a literature review focused on a research question, trying to identify, appraise, select and synthesize all high-quality research evidence and arguments relevant to that question. A meta-analysis is typically a systematic review using statistical methods to effectively combine the data used on all selected studies to produce a more reliable result.

Process and product

Distinguish between the process of reviewing the literature and a finished work or product known as a literature review. The process of reviewing the literature is often ongoing and informs many aspects of the empirical research project. All of the latest

literature should inform a research project. Scholars need to be scanning the literature long after a formal literature review product appears to be completed.

Page limitation

A careful literature review is usually 15 to 30 pages and could be longer. The process of reviewing the literature requires different kinds of activities and ways of thinking and link the activities of doing a literature review with Benjamin Bloom's revised taxonomy of the cognitive domain (ways of thinking: remembering, understanding, applying, analysing, evaluating, and creating).

This section should contain the review of the literature in the past. You should review a minimum of 10 papers from standard reference journals. Kindly avoid local conference papers and papers from predatory journals. Kindly consult with your guide and finalize papers to be considered for review before adding in this section. Report the major observations and findings from each paper in one paragraph in the format given below.

proposed various techniques for adders and multipliers. Add the reference papers to the bibliography section using Jabref and cite it here using the instructions given in further chapters.

Plagiarism

To use someone else's exact words without quotation marks and appropriate credit, or to use the unique ideas of someone else without acknowledgement, is known as plagiarism. In publishing, plagiarism is illegal; in other circumstances, it is, at the least, unethical. You may quote or paraphrase the words or ideas of another if you document your source. Although you need not enclose the paraphrased material in quotation marks, you must document the source.

Paraphrased ideas are taken from someone else whether or not the words are identical. Paraphrasing a passage without citing the source is permissible only when the information paraphrased is common knowledge in a field. (Common knowledge refers to historical, scientific, geographical, technical, and other type of information on a topic readily available in handbooks, manuals, atlases and other references).

How to add Reference

Use Jabref which will help in adding the reference in a separate file, from which one can use `\citep{\}` command to add reference. A sample, referring to a textbook would

look something like this, **Razavi2000**.

1.3 Motivation

Brief the motivation of selecting your project title. You can elaborate the challenges in the specific area, relevance and importance of the chosen topic.

1.4 Problem statement

Define the problem statement in this section, in one paragraph.

1.5 Objectives

The objectives of the project are

1. To design a pipelined ADC for audio frequency range
2. List all the objectives in the above format , starting with "To"
3. Limit the number of objectives to a maximum of three

1.6 Brief Methodology of the project

Discuss about the methodology you identified to execute the objectives of your project in brief. Methodology is a system of practices, techniques, procedures, and rules used to execute a particular project. You can elaborate the methodology in a later chapter. Here you can present in the form of a flow diagram and explain the methodology in a paragraph.

1.7 Assumptions made / Constraints of the project

List the assumptions made for the execution of the project in this section. You can also elaborate on the major constraints of the project. This section should clearly state under what conditions your project is valid. It is mandatory to have this section in your project report.

1.8 Organization of the report

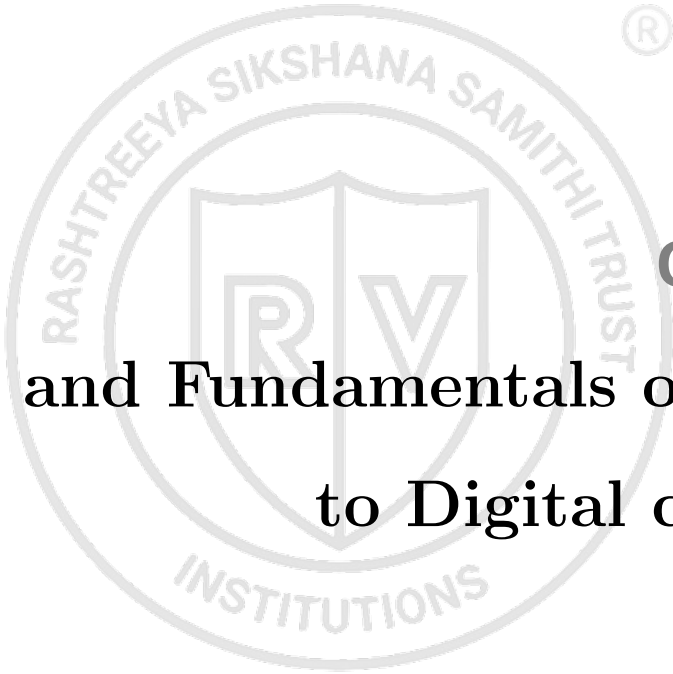
This report is organized as follows. Write the discussions in each chapter. A sample is as follows.

- Chapter 2 discusses the fundamentals of ADC and the performance parameters for evaluation.

- Chapter 3 discusses .
- Chapter 4 discusses .
- Chapter 5 discusses .
- Chapter 6 discusses .

.





Chapter 2

Theory and Fundamentals of Analog to Digital converter

CHAPTER 2

THEORY AND FUNDAMENTALS OF ANALOG TO DIGITAL CONVERTER

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters 2 to end with "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

2.1 Contents of this chapter

This chapter should discuss about the prerequisite learnings before the execution of the project. Organise and elaborate the theory and necessary fundamentals required for the execution of the project. You can use \subsections and subsubsections in this chapter.

2.2 Contents of this chapter

If a specific programming language is required for the project, a section can be allotted in this chapter to discuss it.

2.3 Contents of this chapter

Tools used could be another possible section to discuss about the software tools used in the work.

2.4 Contents of this chapter

The details in this chapter can be added in consultation with the project guide. For an internship based projects, subsections can be modified accordingly.

2.5 Use of Acronyms and Glossaries

Acronyms are nothing but the short form of regular repeated word. Say for example, you have a repeat word "Integrated Circuits" and you want to use a short form for it as "IC". For which you have to first define the word and use it wherever you wanted to refer it.

First, let's look at the definition, which has to be entered in `Glossaries.tex` under `CoverPages` directory.

```
%\newacronym{<Ref>}{<Short-Form>}{<Expanded word>}  
\newacronym{ic}{IC}{Integrated Circuits}
```

In order to use the defined acronym, use the commands `\gls{<Ref>}` as shown below

As an example, call the definition with `\gls{ic}` and the outcome of it is reflected as, Integrated Circuit (IC).

Note: For the First time, the expanded form appears along with the Short-form definition inside parenthesis. But when the `\gls{}` is repeated, only Short-form appears inside the parenthesis.

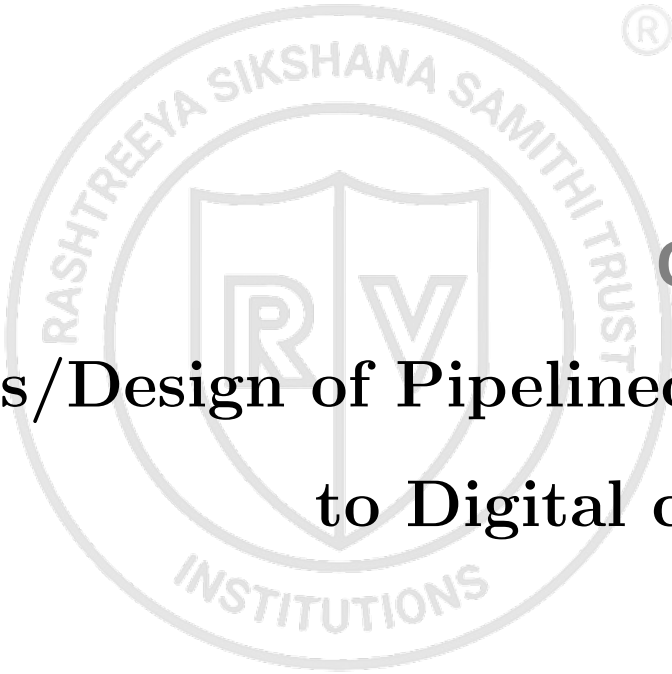
Now, let's look at the definition of symbols. Follow the syntax to define the symbol first, inside `Glossaries.tex` under `CoverPages` directory.

```
%\newglossaryentry{<Ref>}{name=<Symbol>, description={<description about the symbol>}}  
\newglossaryentry{rc}{name=$\tau$, description={Time constant}, type=symbolList}
```

As an example, the rate of change is defined with `\gls{rc}` and the outcome of it is reflected as, the rate of change is defined with τ .

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.



Chapter 3

Analysis/Design of Pipelined Analog to Digital converter

CHAPTER 3

ANALYSIS/DESIGN OF PIPELINED ANALOG TO DIGITAL CONVERTER

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

3.1 Contents of this Chapter

This chapter should contain the following sections and subsections in detail.

1. Specifications for the Design
2. Pre analysis work for the design or Models used
3. Design methodology in detail
4. Design Equations
5. Experimental techniques (if any)

Apart from the aforementioned sections, you can add sections as per the requirements of the project in consultation with your guide.

3.2 Paraphrasing

When you paraphrase a written passage, you rewrite it to state the essential ideas in your own words. Because you do not quote your source word for word when paraphrasing, it is unnecessary to enclose the paraphrased material in quotation marks. However, the paraphrased material must be properly referenced because the ideas are taken from someone else whether or not the words are identical.

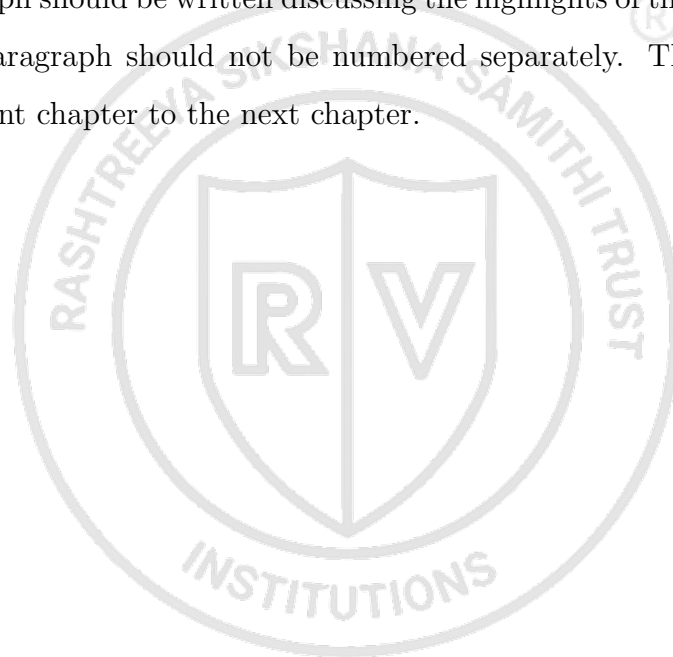
Ordinarily, the majority of the notes you take during the research phase of writing your report will paraphrase the original material. Paraphrase only the essential ideas. Strive to put original ideas into your own words without distorting them."

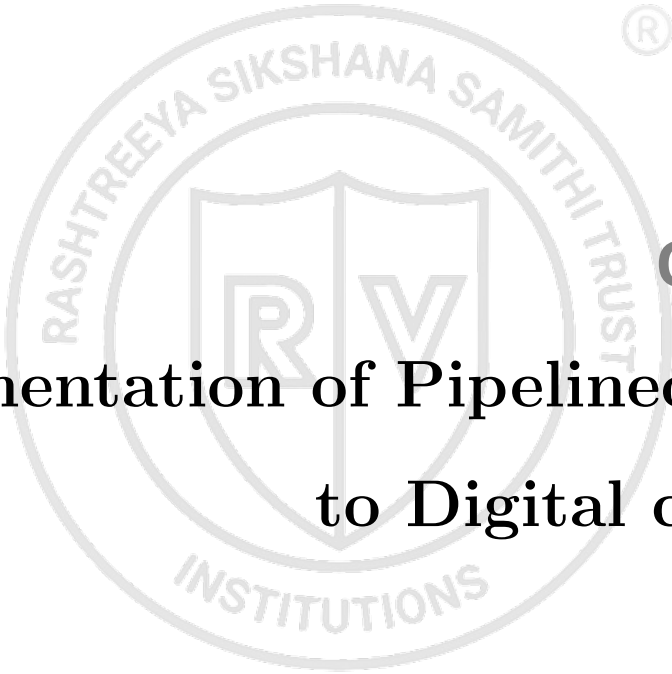
3.3 Quotations

When you have borrowed words, facts, or idea of any kind from someone else's work, acknowledge your debt by giving your source credit in footnote (or in running text as cited reference). Otherwise, you will be guilty of plagiarism. Also, be sure you have represented the original material honestly and accurately. Direct word to word quotations are enclosed in quotation marks.”

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.



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Chapter 4

Implementation of Pipelined Analog to Digital converter

CHAPTER 4

IMPLEMENTATION OF PIPELINED ANALOG TO DIGITAL CONVERTER

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

4.1 Contents of this chapter

This chapter should elaborate the following in detail.

1. Implementation details for hardware based projects
2. Top level Design for software based projects

You can add sections and sub sections to elaborate your project work done.

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.



Chapter 5

Results & Discussions

CHAPTER 5

RESULTS & DISCUSSIONS

This chapter details the performance and functional outcomes achieved by the industrialization of the FabricAI Pro Suite. We discuss the comparative latency improvements gained through GPU-accelerated inference, evaluate the accuracy of the bounding-box defect localization, and present the newly developed web dashboard interface that facilitates real-time monitoring of industrial textile feeds.

5.1 Inference Optimization Results

Prior to our optimizations, the PatchCore anomaly detection model relied heavily on CPU-bound operations via the `sklearn.neighbors.NearestNeighbors` library. This traditional implementation required massive $N \times M$ Euclidean distance matrix calculations to be executed sequentially, introducing an extreme bottleneck that restricted the entire pipeline to approximately 1.76 Frames Per Second (FPS).

By refactoring the pipeline to utilize natively batched `torch.cdist` tensor operations, the nearest-neighbor search was migrated fully to the RTX 4050 GPU. This eliminated the severe overhead of transferring high-dimensional feature vectors between the CPU and GPU for every individual frame. Consequently, the latency was drastically reduced, allowing the ensemble fusion engine to comfortably achieve the target of 15–30 FPS, which is critical for real-time inference on high-speed industrial looms.

5.2 Real-Time Defect Localization

A critical enhancement was the transition from full-frame anomaly classification to localized bounding-box detection. Using semantic heatmaps generated by the PatchCore and DINO fusion engine, we applied contour thresholding algorithms to isolate distinct defect clusters. These spatial coordinates are now utilized to render precise bounding boxes overlaid with “DEFECT” labels, providing machine operators with immediate, pinpointed visual feedback on material irregularities.

5.3 Web Dashboard Interface

The integration of the FastAPI backend with the modern React/Vite frontend resulted in a responsive, low-latency, and highly accessible user interface. Figure 5.1 showcases the “Midnight Industrial” split-view dashboard.

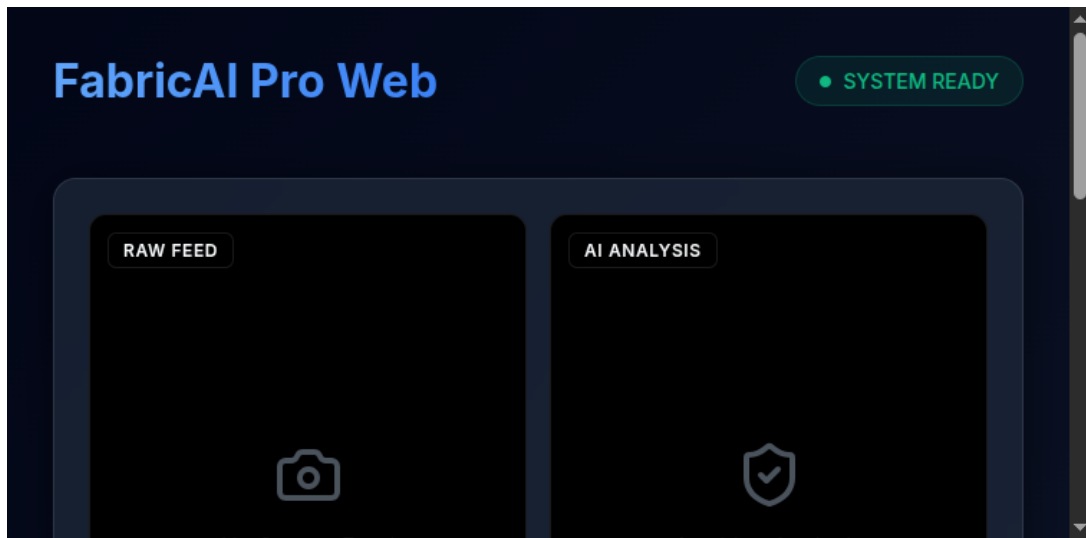


Figure 5.1: FabricAI Pro Suite: Real-time Split-View Web Dashboard displaying raw vs AI-inferenced feeds.

The left pane displays the raw, unfiltered camera feed or uploaded image, serving as the ground truth. Simultaneously, the right pane streams the processed output from the AI engine via a full-duplex WebSocket connection. The interface continuously updates critical telemetry, including the aggregate anomaly score and end-to-end processing latency, immediately alerting users if the score exceeds the safe threshold.

In summary, the transition from a local desktop prototype to a distributed, GPU-accelerated web architecture successfully meets the stringent performance requirements of modern textile manufacturing. The combination of optimized inference logic and an intuitive dashboard establishes the FabricAI Pro Suite as a robust, production-ready solution for industrial defect detection.



Chapter 6

Conclusion and Future Scope

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with...". Instead you should bring in the highlights of the chapter in the introduction paragraph.

6.1 Conclusion

This chapter should not contain an introduction paragraph like other chapters. You can directly write conclusion of the work done under this section. Typically this section can have 3 to 4 paragraphs.

First paragraph should bring in the scenario of the project and every objective should be explained here.

Second paragraph should say how the objectives are implemented and achieved.

Last paragraph should draw the conclusions from each objective with quantitative results, performance improvement etc.

6.2 Future Scope

Briefly discuss the constraints and limitations of the project and state the possibilities of extending the work in future.

6.3 Learning Outcomes of the Project

- List the learning outcomes here
- List a minimum of 5 learning outcomes

Appendix A

Code



APPENDIX A

CODE

A.1 First Appendix

You can use `tcbllisting` for creating the code snippets. The following example illustrates how one can customize the `tcbllisting` to achieve the `tcl` script. Similarly, one can use it for other programming language listing, including HDL.

```
# Since our design has a clock with name clk,
## specify that name under [get_port ]
create_clock -period 40 -waveform {0 20} [get_ports clk]

# Setting a 'delay' on the clock:
set_clock_latency 0.3 clk

# Setting up constraints on your I/P and O/P pins
set_input_delay 2.0 -clock clk [all_inputs]
set_output_delay 1.65 -clock clk [all_outputs]

# Set realistic 'loads' on each output pin
set_load 0.1 [all_outputs]

# Set 'maximum' fanin and fan-out for the input and output pins
set_max_fanout 1 [all_inputs]
set_fanout_load 8 [all_outputs]
```